DECISION MAKING & BRANCHING IN C

Subject: Programming for Problem Solving – I

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Control Structures

- Control structures are used for decision making in any programming language.
- Most programming languages along with C supports following control structures.
  - **Sequence**
  - **Decision**
  - **Looping**

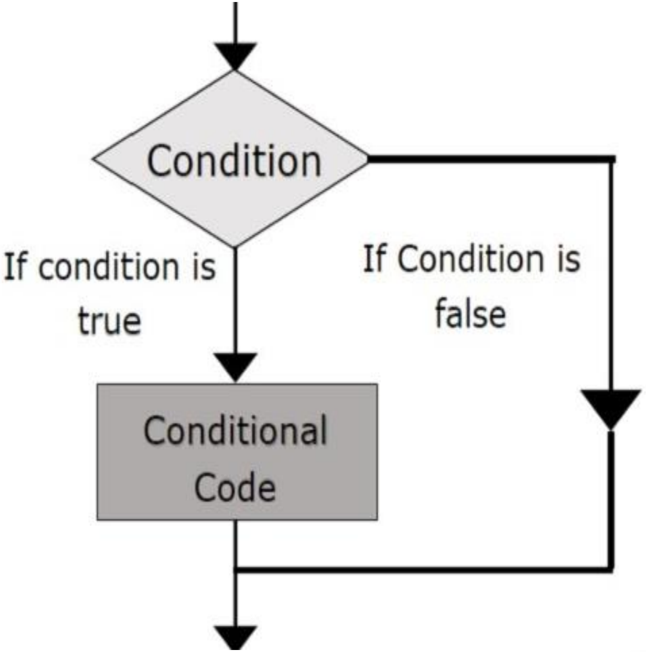
Decision Making in C

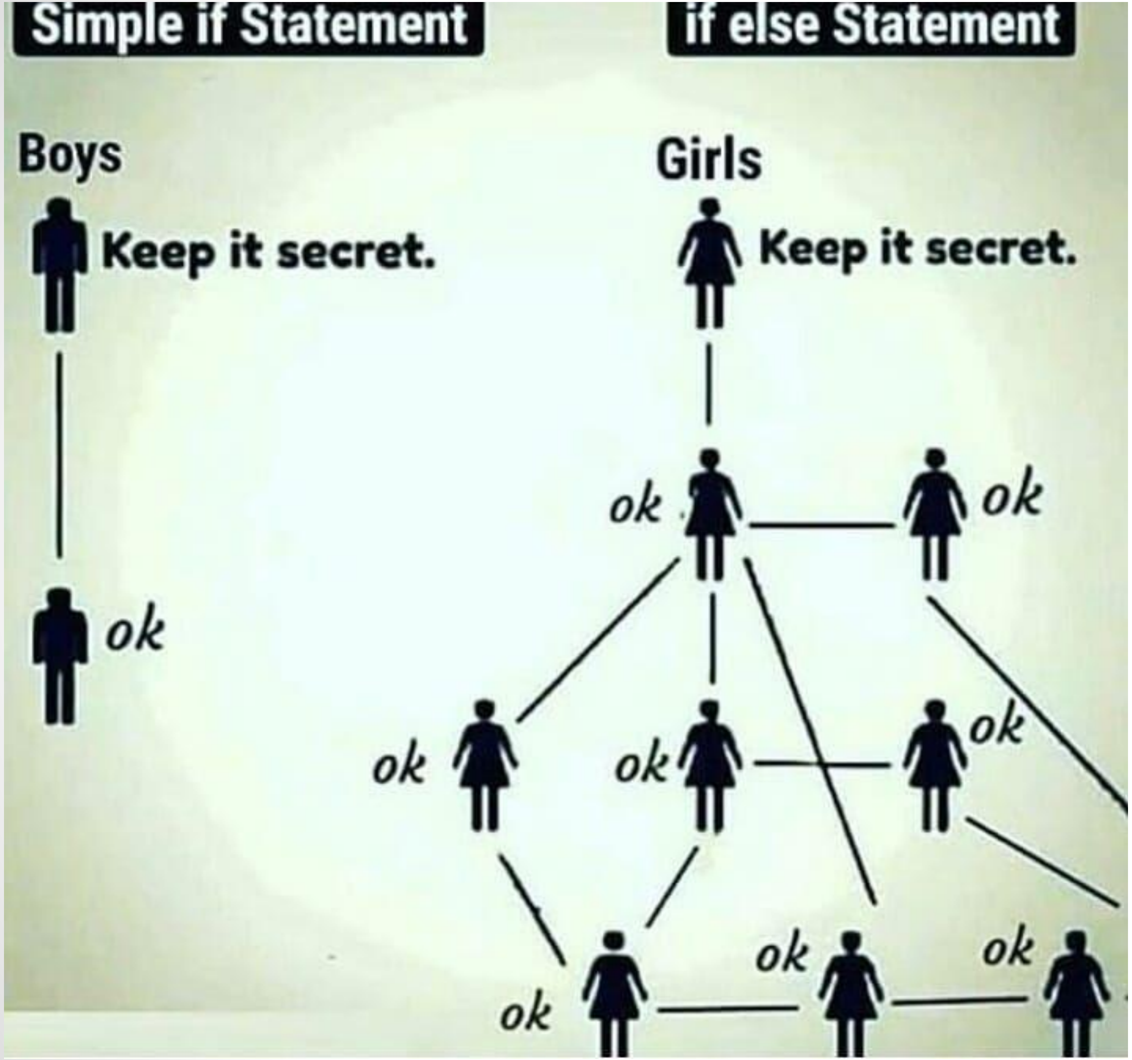
- C supports following decision making statements:
  - **if-else**
  - **multiple if-else (ladder if-else)**
  - **switch**
  - **goto**
- All programming or real world problems are not always sequential like finding area of circle or printing Fibonacci series.
- Hence, decision making ability is important part of any language.

if statement

- It checks for a condition in program.
- It executes set of statements if condition is true. otherwise it continues with the flow of program.
- Syntax:

```
if (condition)
{
        set of statements;
}
```





if statement
vs
if-else statement

if-else statements

- if (condition)

        statement1;

  else

        statement2;

- if (condition)

  {

        Block1

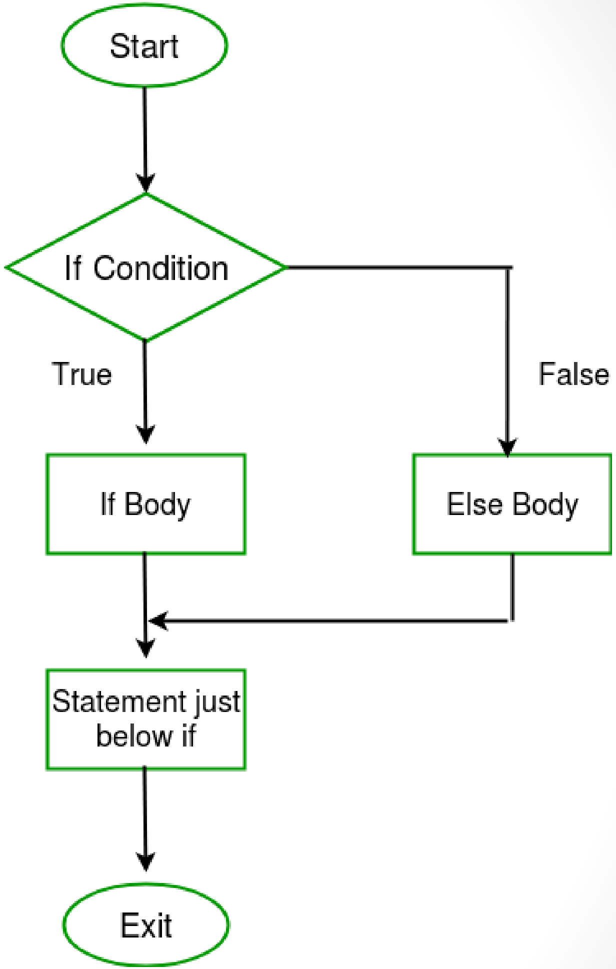
  }

  else

  {

        Block2

  }



Some examples of if-else

1. ```
   if (age >= 18)
       printf("You  are eligible to apply for voter id.");
   else
       printf("You  are minor.");
   ```
2. ```
   if (x * y == 0)
       printf("Either x or y is zero.");
   else
       printf("Both x and y are non-zero numbers.");
   ```

Nested if-else

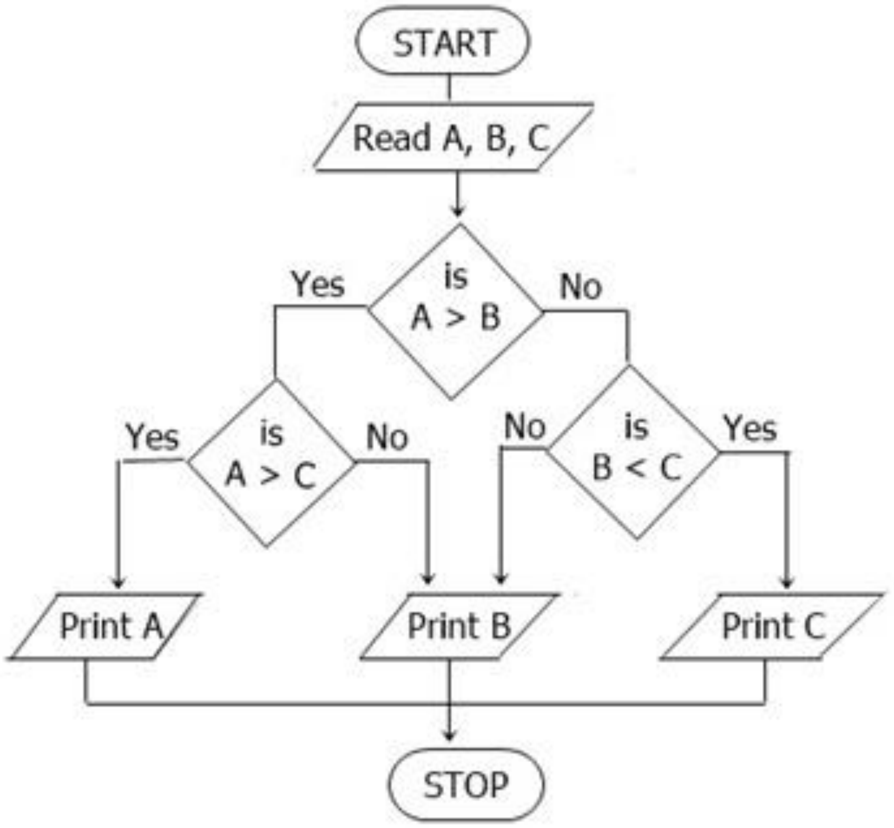
- When we use if-else statements inside already existing if-else statements, it is known as nested if-else.
- Any level or degree of nesting is possible in if-else statements. Just make sure to follow correct syntax.

Syntax

```
if ( condition1 )
{
    if ( condition2 )
    {
        ....
        True block of statements 1;
    }
    ....
}
else
{
    False block of condition1;
}
```

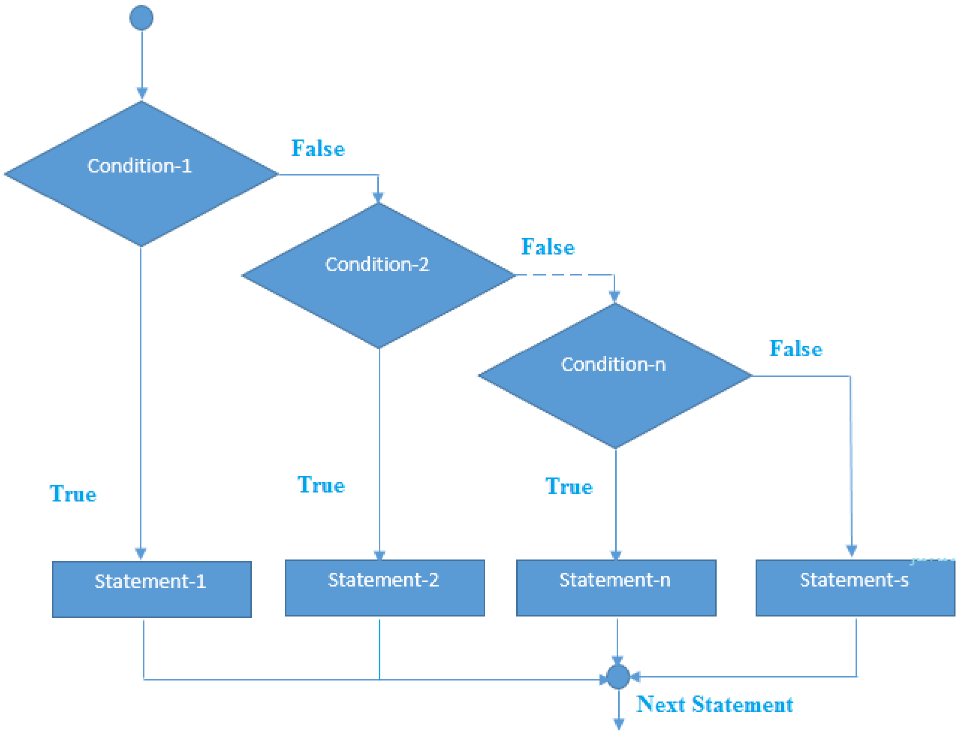
Nested if-else Example

```
if(a>b)
{
    if(a>c)
        printf("A is largest.");
    else
        printf("C is largest.");
}
else
{
    if(b>c)
        printf("B is largest.");
    else
        printf("C is largest.");
}
```



Multiple if-else (ladder if-else)

```
if(condition1)
    statement1;
else if(condition2)
    statement2;
else if(condition3)
    statement3;
.
.
.
else if(conditionN)
    statementN;
else
    Default_Statement;
```



Multiple if-else Example

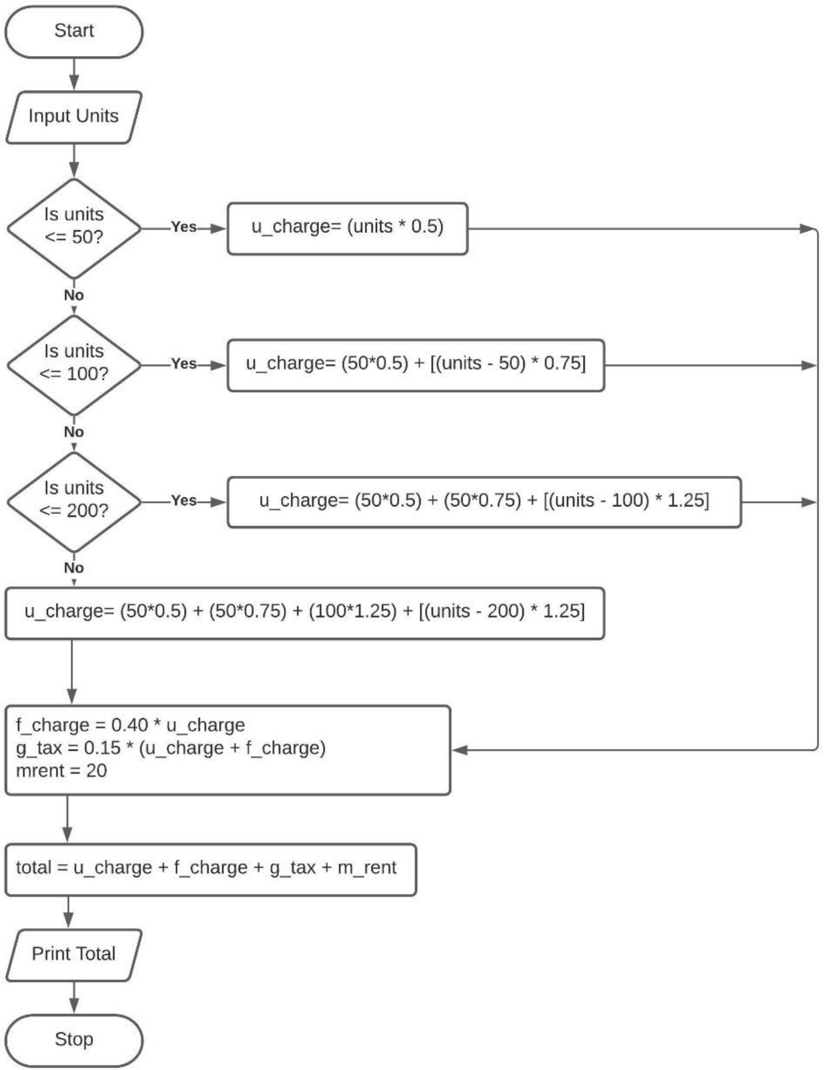
Calculate Electricity Bill:

A. Given the number of units, unit charges are as follow:
   First 50 units <- 0.50rs / unit
   Next 50 units <- 0.75rs / unit
   Next 100 units <- 1.25rs / unit
   Above 200 units <- 1.75rs / unit
B. Fuel Surcharge is 40% of unit charges
C. Govt. Tax is 15% of unit and fuel charges
D. Meter rent is fixed at 20rs

- Total bill is sum of A, B, C and D.

Calculate Electricity Bill

Flowchart



Switch Statement

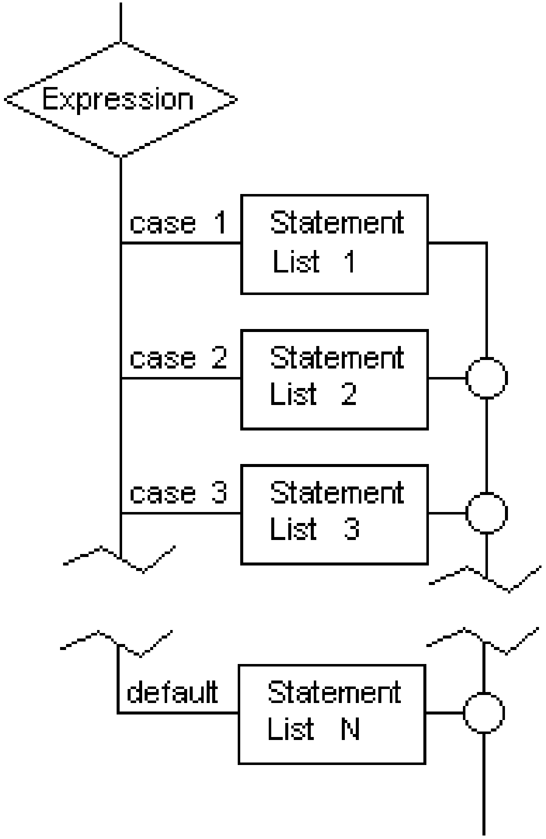
- Multi-choice statements, where block of correct choice is execute.
- Resembles with if-else ladder, but only difference is that it will jump directly to correct condition rather than checking the conditions one-by-one.
- Switch statement have variable or expression as decision making.

People asking me if a if statement and switch is the same

Well yes

Switch Syntax / Flowchart

```
switch(variable or expression)
{
    case lable1:
        Block1 of statements;
        break;
    case lable2:
        Block2 of statements;
        break;
    .
    .
    .
    case lableN:
        BlockN of statements;
        break;
    default:
        Default block of statement
}
```



Simple Calculator Program using Switch Statement

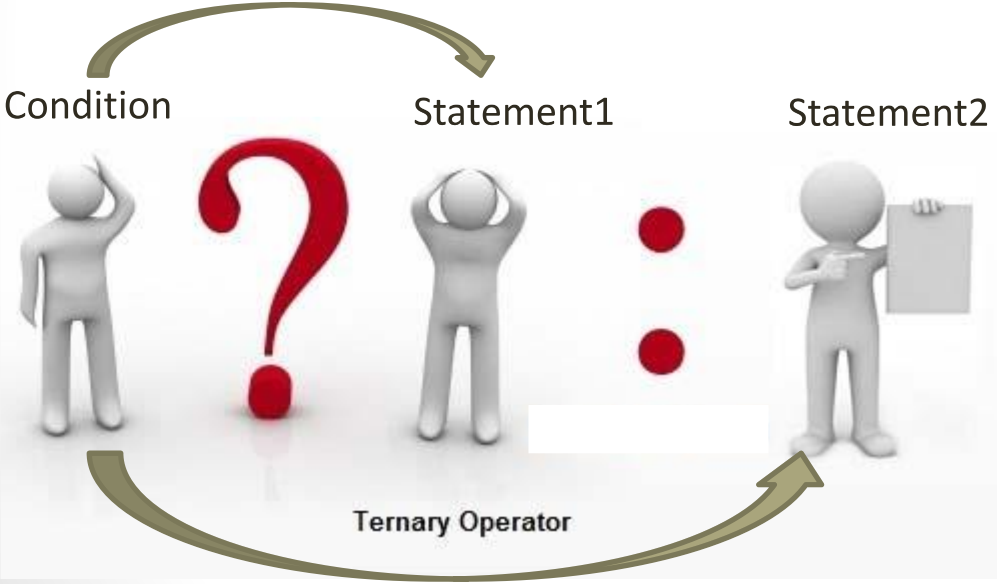
- Switch to editor

Ternary (?:) Operator

- If any operator is used on three operands or variable is known as **Ternary Operator**.
- It can be represented with ’**? :’** . It is also called as **conditional operator**.
- The ternary operator is an operator that takes three arguments.
  - The first argument is a comparison argument,
  - the second is the result upon a true comparison, and
  - the third is the result upon a false comparison.

Use of Ternary (?:) Operator

- Ternary operator is shortened way of writing an if-else statement.
- E.g: condition ? Statement1 : statement2

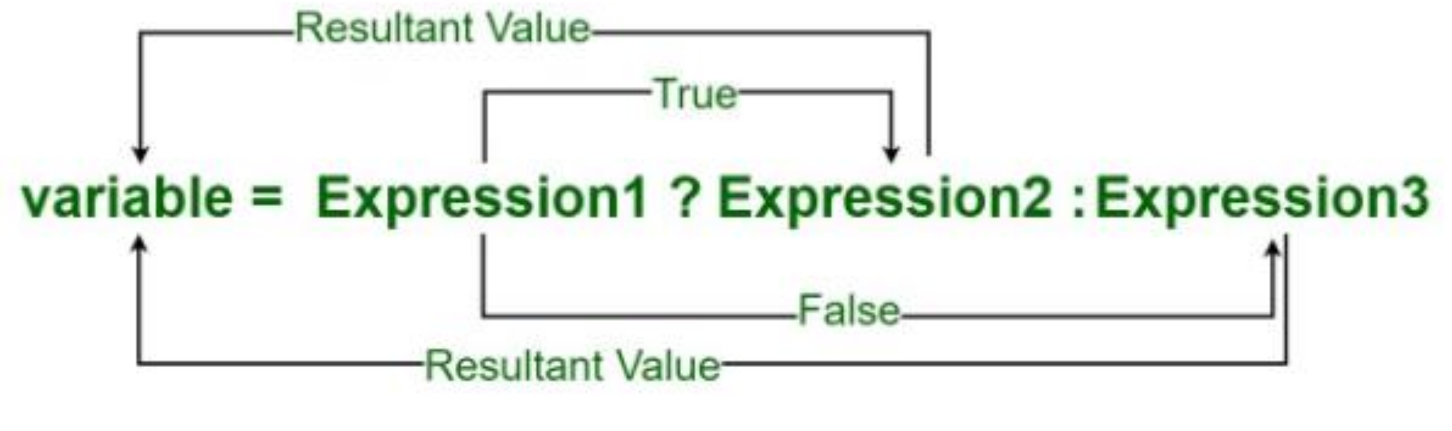


- **if-else syntax:**

```
if (condition)
        statement1;
else
        statement2;
```

E.g. of Ternary (?:) Operator

1. a < b ? printf("a is less") : printf("a is greater");
2. min = a < b ? a : b;
3. large = a > b ? (a > c ? a : c) : (b > c ? b : c);



goto Statement

- Used to transfer the control of program unconditionally from one part to another part.
- Structured language is supposed to be predictable in terms of flow of program.
- Hence, good programmers suggest to avoid use of **goto** statements in C program.
- goto statements use lable and **:** (colon) sign in its syntax.
- lable is a valid identifier.

goto Statement

Syntax:

```
Syntax1      |   Syntax2
------------------------------
goto label;  |    label:
.            |    .
.            |    .
.            |    .
label:       |    goto label;
```

goto Statement Example

```
int marks;

label1:
printf("Enter your marks: ");
scanf("%d",&marks);

if(marks<0 || marks>100)
{
    printf("You have entered incorrect value of marks. Try again...\n");
    goto label1;
}
else
{
    printf("You have entered correct marks.");
}
```